

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018, KARNATAKA



Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**PRE-FINAL YEAR
(2025 – 2026)**

MULTIMEDIA PROJECT BASED LEARNING

On

"CLOUD BASED IMAGE SURVEILLANCE"

Subject Faculty

Dr. Krupa Rasane
Professor , Dept. of E&CE
JCE, Belagavi

PROJECT MEMBERS

NAME- Siddharth Kshatriya	USN-2JI23EC140
NAME- Siddeshwar Kumbar	USN-2JI23EC139
NAME- Sujal Patil	USN-2JI23EC156

Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI
T.S.NAGAR, MACHHE – 590014, KARNATAKA



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

CERTIFICATE

This is to certify that the **Multimedia Communication Project** entitled “**CLOUD BASED IMAGE SURVEILLANCE**” carried out by **Siddharth K. (2JI23EC140), Siddeshwar K. (2JI23EC139) and Sujal P. (2JI23EC156)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project Multimedia based learning report has been approved as it satisfies the academic requirement in respect of CIE prescribed for subject Multimedia.

Demonstration	Report	Presentation	Total
5 marks	5 marks	5 marks	15 marks

Dr.Krupa Rasane

Project Guide

Dept. of E&CE, JCE Belagavi

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INTRODUCTION

Multimedia communication plays a crucial role in modern embedded and IoT systems, enabling the transmission of images, audio, and video data over networks for real-time monitoring and analysis. With the rapid growth of smart surveillance and automation systems [4], low-cost and efficient solutions for capturing and transmitting visual data have become highly important.

This project focuses on the design and implementation of a simple motion detection and image transmission system using the ESP32-CAM module. The system utilizes a PIR (Passive Infrared) sensor to detect motion in the environment. Upon detecting movement, the ESP32-CAM captures an image and transmits it to a cloud platform (Firebase), allowing remote access and storage.

The integration of embedded hardware with cloud services demonstrates a practical application of multimedia communication, where image data is captured, processed, and transmitted over the internet [1]. Additionally, a basic analysis is performed to study how different image resolutions affect parameters such as file size, upload time, delay, and visual quality.

This project highlights the feasibility of implementing a compact, cost-effective, and efficient smart monitoring system using widely available components and cloud infrastructure.

OBJECTIVES

- To design and implement a motion detection system using a PIR sensor.
- To interface the ESP32-CAM module for capturing images upon motion detection.
- To transmit captured images to a cloud platform (Firebase) using internet connectivity.
- To enable remote storage and access of multimedia (image) data.
- To analyze the impact of different image resolutions on system performance.
- To compare parameters such as file size, upload time, total delay, and visual quality.
- To develop a low-cost and efficient multimedia communication system for basic surveillance applications.

COMPONENTS / TOOLS USED

Hardware Components:

- **ESP32-CAM Module:**

The ESP32-CAM is a compact microcontroller module based on the ESP32 with an integrated OV2640 camera. It supports Wi-Fi and Bluetooth connectivity, making it suitable for IoT and multimedia applications. It has limited GPIO pins, built-in PSRAM for image processing, and supports multiple image resolutions including QVGA, VGA, and SVGA [2].



- **HC-SR501 PIR Sensor:**

The HC-SR501 is a passive infrared (PIR) sensor used for motion detection. It detects changes in infrared radiation caused by human or object movement. It operates typically at 5V, has adjustable sensitivity and delay time, and provides a digital output signal when motion is detected.



- **ESP32-CAM MB Base Board:**

The ESP32-CAM MB is a programmer base board designed for easy interfacing with the ESP32-CAM module. It includes a built-in USB-to-serial converter, voltage regulator, and micro-USB port, allowing direct programming and power supply without the need for an external FTDI module.



- **Jumper Wires** –For making electrical connections between components
- **Power Supply (5V)** – Provides required power to the ESP32-CAM and sensor. Here we are using ARDUINO UNO R3 as a power supply for a stable 5v input. A dedicated power supply is recommended.

Software Tools:

- **Arduino IDE:**
A development environment used to write, compile, and upload code to the ESP32-CAM.
- **Firebase (Google Firebase):**
A cloud platform used to store and manage the captured images.
- **ESP32 Board Package:**
Provides necessary libraries and board support to program ESP32 devices within the Arduino IDE

Other Requirements:

- **Wi-Fi Network** – Enables internet connectivity for uploading images to Firebase.

CODE

The system is programmed using the Arduino IDE for the ESP32-CAM module. The code was developed with the help of ESP32 board packages, required libraries, and assistance from ChatGPT for structuring and debugging.

LIBRARY INCLUSION AND CONFIGURATION

The required libraries for camera operation, Wi-Fi connectivity, and Firebase communication are included at the beginning. Firebase credentials and Wi-Fi details are also defined.

```
<> C++  
  
#include "esp_camera.h"  
#include <WiFi.h>  
#include <Firebase_ESP_Client.h>
```

These libraries enable image capture, internet connectivity, and cloud communication respectively.

CAMERA INITIALIZATION

The ESP32-CAM is configured by assigning GPIO pins and setting parameters such as resolution and image quality.

```
<> C++  
  
cam_config.frame_size = FRAMESIZE_VGA;    // 640x480  
cam_config.jpeg_quality = 12;  
cam_config.fb_count = 2;
```

This configuration provides a balance between image quality and transmission speed.

WI-FI AND FIREBASE SETUP

The ESP32 connects to a Wi-Fi network and initializes Firebase using API credentials.

```
</> C++  
WiFi.begin(ssid, password);  
  
Firebase.begin(&config, &auth);  
Firebase.reconnectWiFi(true);
```

The system waits until authentication is complete before proceeding.

MOTION DETECTION LOGIC

The PIR sensor continuously monitors motion. When motion is detected, the system triggers image capture.

```
</> C++  
  
int pirState = digitalRead(PIR_PIN);  
  
if (pirState == HIGH && millis() - lastTriggerTime > cooldown)
```

A cooldown delay is implemented to avoid repeated triggering

IMAGE UPLOAD TO FIREBASE

The captured image is uploaded to Firebase Storage with a unique filename.

```
</> C++  
  
Firebase.Storage.upload(  
    &fbdo,  
    STORAGE_BUCKET_ID,  
    fb->buf,  
    fb->len,  
    fileName,  
    "image/jpeg"  
);
```

SUPPORTING FILES

In addition to the main file (`CameraWebServer.ino`), supporting files such as `app_httpd.cpp`, `camera_pins.h`, `camera_index.h`, and `board_config.h` are used for hardware configuration and camera handling.

SUMMARY

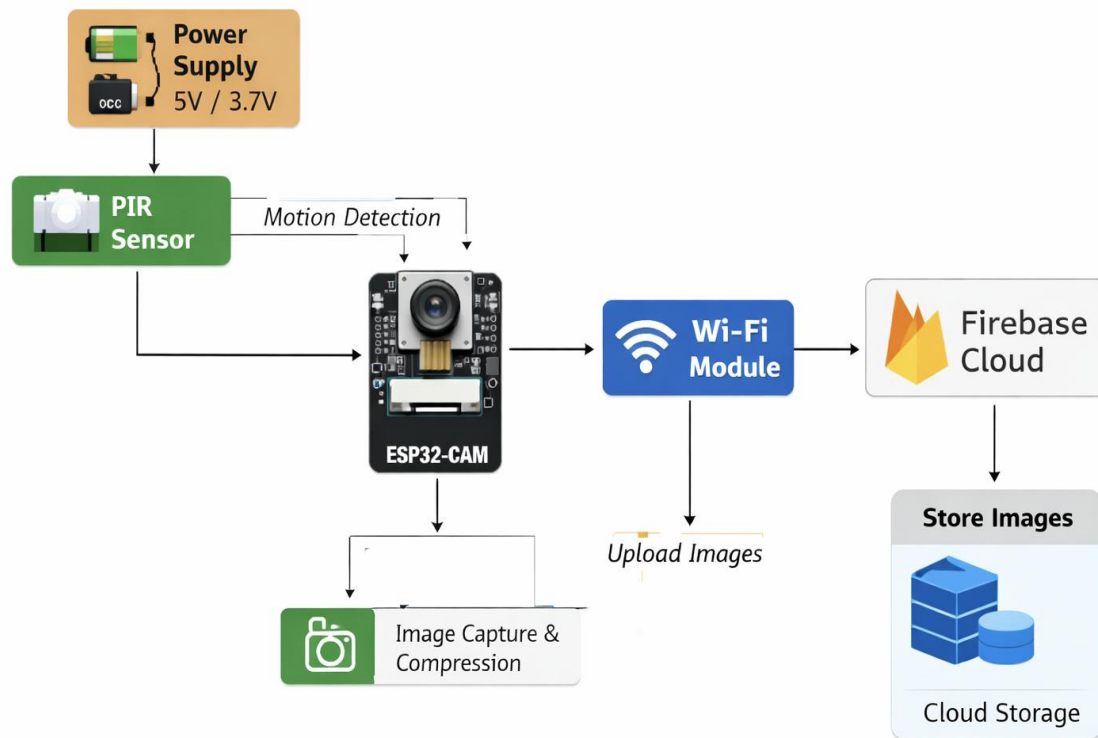
The code integrates motion sensing, image capture, and cloud communication effectively. It demonstrates a simple yet functional multimedia communication system using embedded hardware and cloud services.

WORKING PRINCIPLE

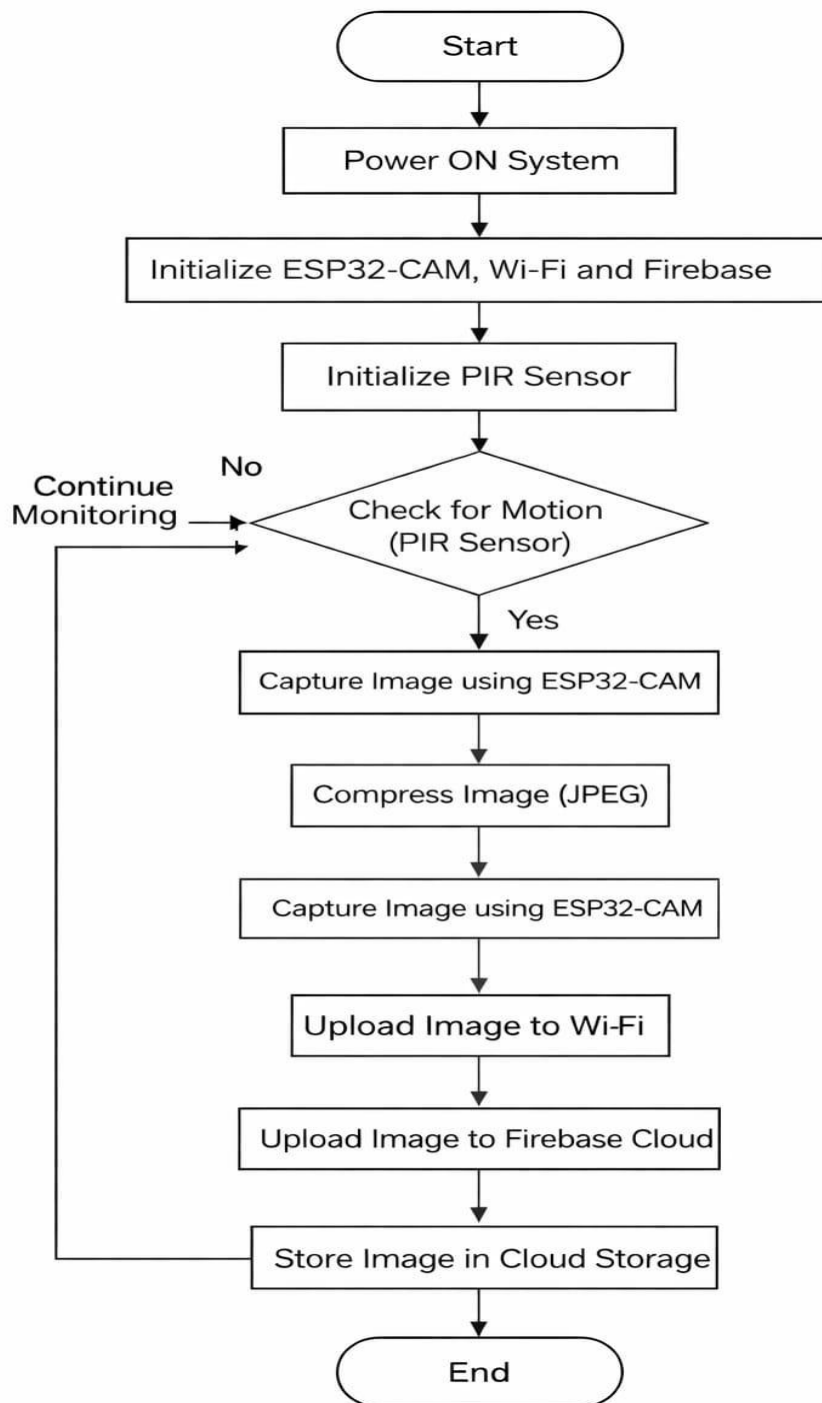
- The ESP32-CAM initializes the camera, connects to Wi-Fi, and authenticates with Firebase on startup.
- The HC-SR501 PIR sensor continuously monitors the surroundings for motion [4].
- When motion is detected, the sensor outputs a HIGH signal to the ESP32-CAM.
- The ESP32-CAM checks a cooldown condition to avoid repeated triggering.
- Upon valid detection, the camera captures an image and stores it in a frame buffer.
- The captured image is uploaded to Firebase Cloud Storage using Wi-Fi with a unique filename [1].
- Parameters such as image size, upload time, and total delay are recorded.
- The uploaded images can be accessed remotely, completing the multimedia communication process.

PROJECT FLOW

BLOCK DIAGRAM:



FLOWCHART:

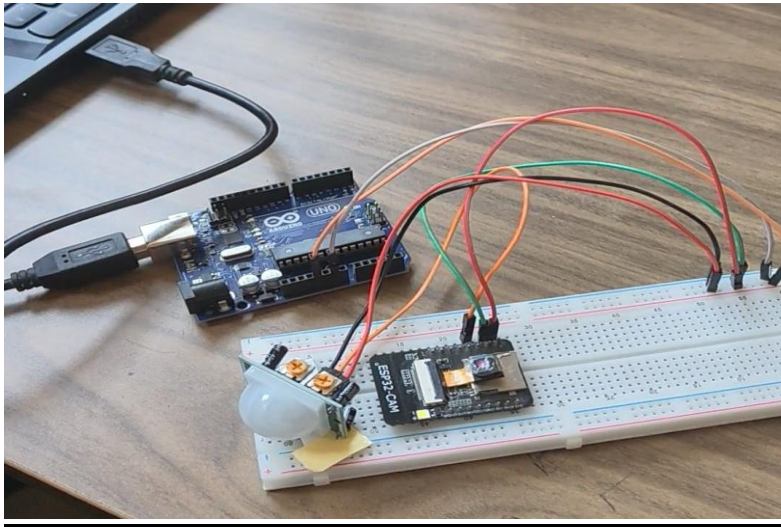


APPLICATIONS

- **Home Security Systems:** Used for detecting intrusions and capturing images for monitoring.
- **Smart Surveillance:** Enables low-cost surveillance in offices, shops, and public spaces.
- **Wildlife Monitoring:** Captures images of animals based on motion in remote areas.
- **Industrial Safety:** Detects unauthorized access or movement in restricted zones.
- **Smart Door Monitoring:** Can be used as a basic door camera to capture visitors.
- **IoT-based Monitoring Systems:** Demonstrates real-time multimedia data transmission in IoT applications [3].

RESULTS

Prototype:

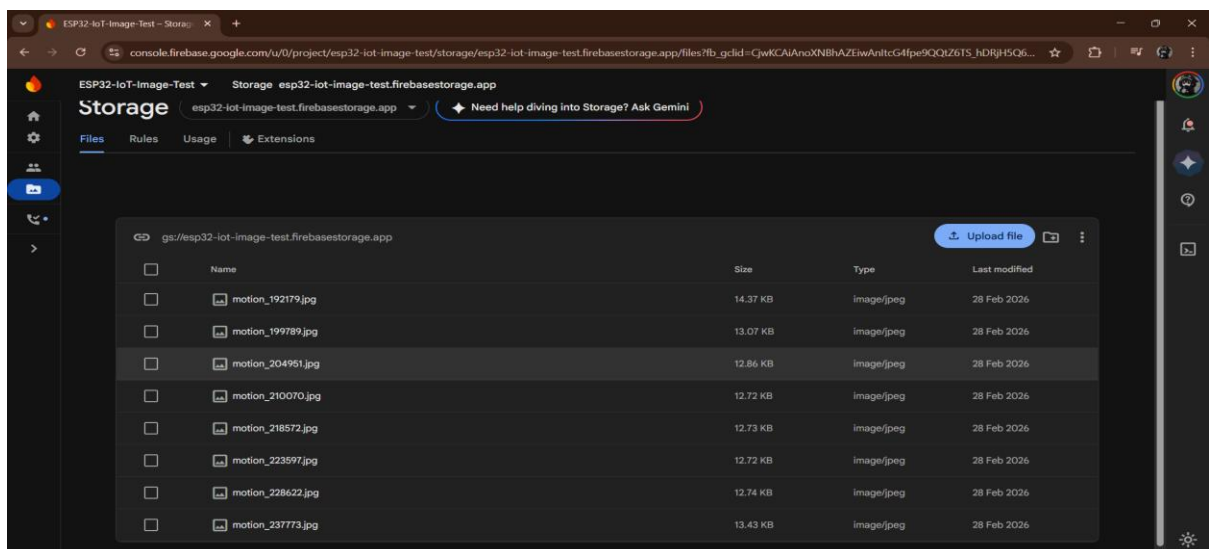


Rate distortion table:

Resolution	Pixel Count	Avg File Size (KB)	Avg Upload Time (ms)	Avg Total Delay (ms)	Visual Quality
QVGA (320×240)	76,800	4.2	520	670	Low
VGA (640×480)	307,200	13.0	1300	1450	Medium
SVGA (800×600)	480,000	21.5	2350	2520	High

This trend is consistent with observations in similar ESP32-based systems [3].

Firestore upload:



CONCLUSION

This project demonstrates a simple and cost-effective multimedia communication system using the ESP32-CAM and PIR sensor. The system successfully detects motion, captures images, and transmits them to Firebase cloud storage for remote access.

The implementation shows how embedded systems can be integrated with cloud platforms to enable real-time data communication. The performance analysis highlights the trade-off between image quality, file size, and transmission delay.

Overall, the project validates the feasibility of developing a basic smart surveillance system using low-cost hardware and widely available software tools.

REFERENCES

- [1] F. Esparza-Segundo, S.-L. Enríquez-Arcadio, J. C. Gonzalez-Islas, A. Suarez-Navarrete, and M. de J. Gutiérrez-Sanchez, “ESP32-CAM-based smart surveillance: Real-time transmission and alerts via Telegram,” *Programación Matemática y Software*, vol. 17, no. 2, pp. 48–60, 2025.
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